IBM Data Analyst Capstone Project

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EXECUTIVE SUMMARY

- Washington DC is the city with the highest job postings.
- Swift is the highest paying programming language with an average salary of $130801.
- JavaScript is top used and desired language. There’s interest in Go and Rust for future projects.
- PostgreSQL leads both current use and future desire.
- React and Node.js dominate the trend bubble chart for future work .
- AWS and Google Cloud are current leaders . Supabase and Vercel appear emerging interest for future work.
- 41.3% of respondents are 25–34 years old .

INTRODUCTION

This report aims to identify and analyze emerging and future skill requirements within the global developer community to ensure the organization remains competitive in a rapidly evolving tech landscape. It is designed for internal stakeholders, including executive leadership, project managers, and the recruitment department, to guide strategic decision-making regarding talent acquisition and training.

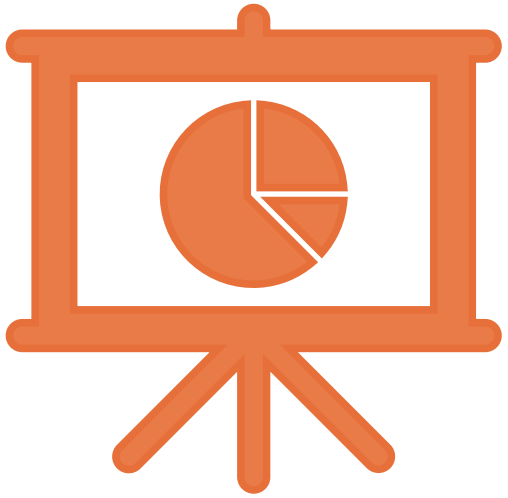
This analysis provides a data-driven roadmap for future technology investments, moving beyond current usage.

- Washington DC has the highest job postings.
- Swift language is the highest paying with the highest average annual salary.
- Identifies that while traditional tools remain industry standards, there is a significant surge in interest for emerging platforms.
- Highlights that the primary workforce is early-to-mid career (ages 25–34) and highly educated, majority of the respondents holding a Bachelor’s degree.

METHODOLOGY

- Information was gathered from multiple primary sources, including the latest Stack Overflow Developer Survey, various job postings, and training portals to ensure a holistic view of the market.
- Collection techniques involved web scraping internet sites for real-time job demand, accessing APIs for structured technical data, and processing large-scale datasets in .csv and Excel formats.
- The raw data underwent preparation, including handling missing values (nulls, normalizing inconsistent technology names, and filtering data to focus on top-tier trends.
- Technologies were segmented into distinct groups such as Programming Languages, Databases, and Web Frameworks to track specific market niches.
- Data was specifically structured to compare "Current Usage" against "Future Trends" to identify predictive growth areas.

Results



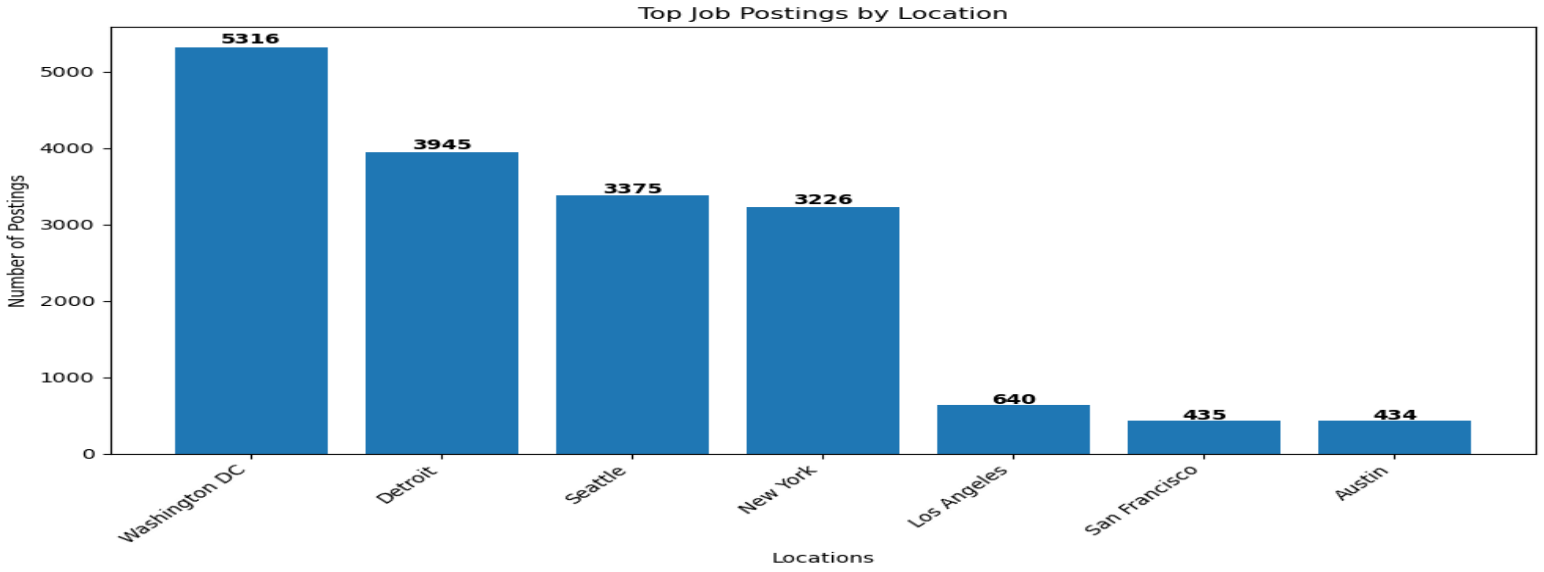
<Please present your results in the following slides.>

JOB POSTINGS

Collecting_job_data_using_ | Jobs_API (1).ipynb

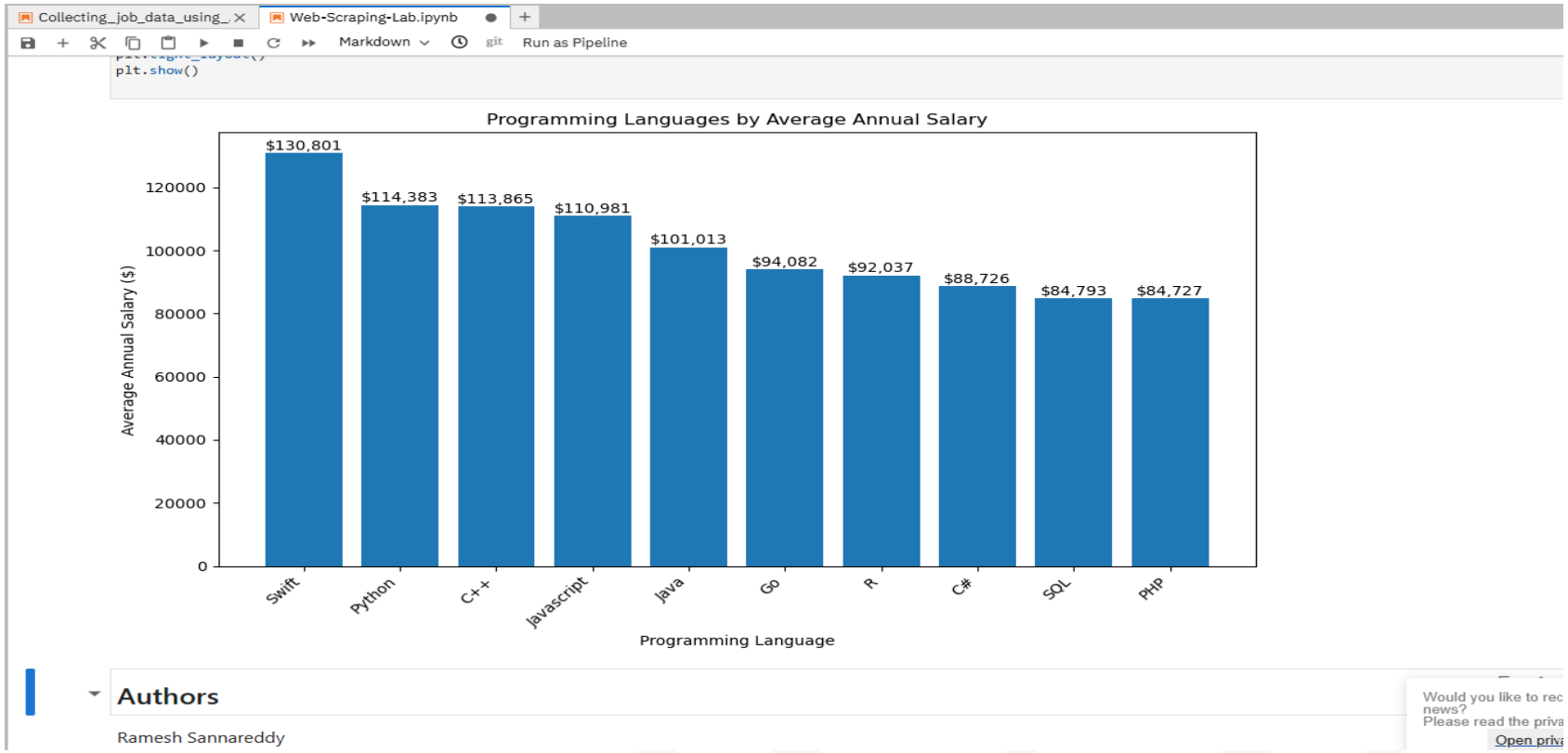
Code | git | Run as Pipeline

```
# Display the sorted data for verification
df_sorted
```



- Washington DC has the highest job postings with Austin having the least.

POPULAR LANGUAGES

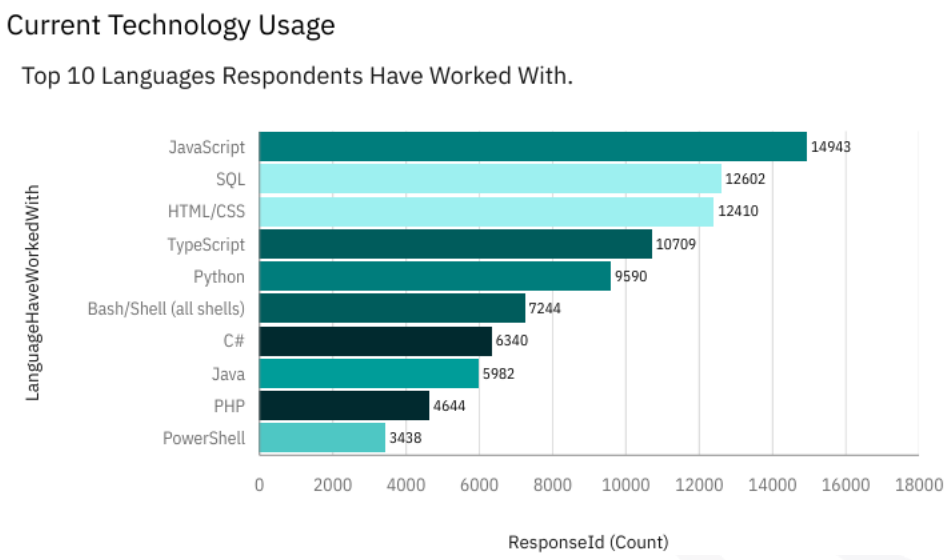


- Swift is the highest paying language with and average of $130801.

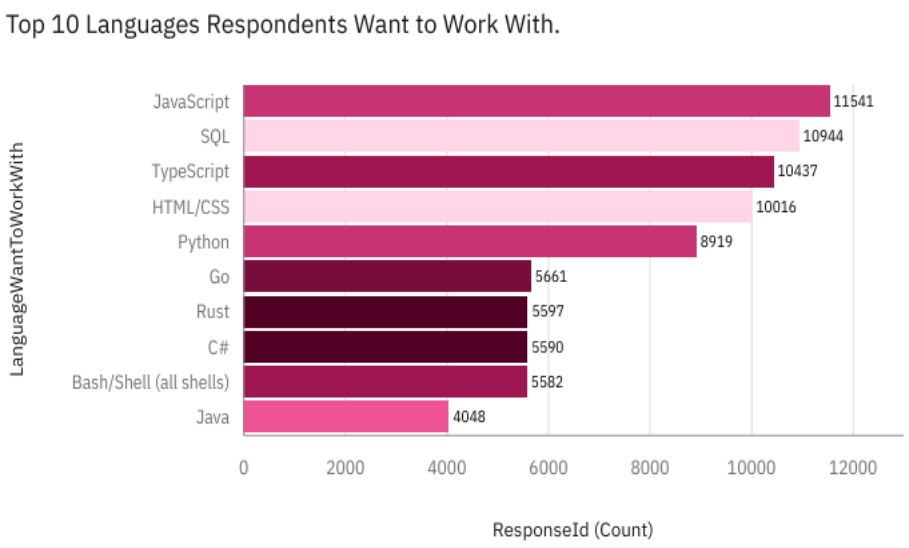
PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript remains the top used (14,943) and desired (11,541) language .
- Go and Rust entered the top 10 for future interest .
- PHP dropped out of the top 10 future languages respondents want to work with.

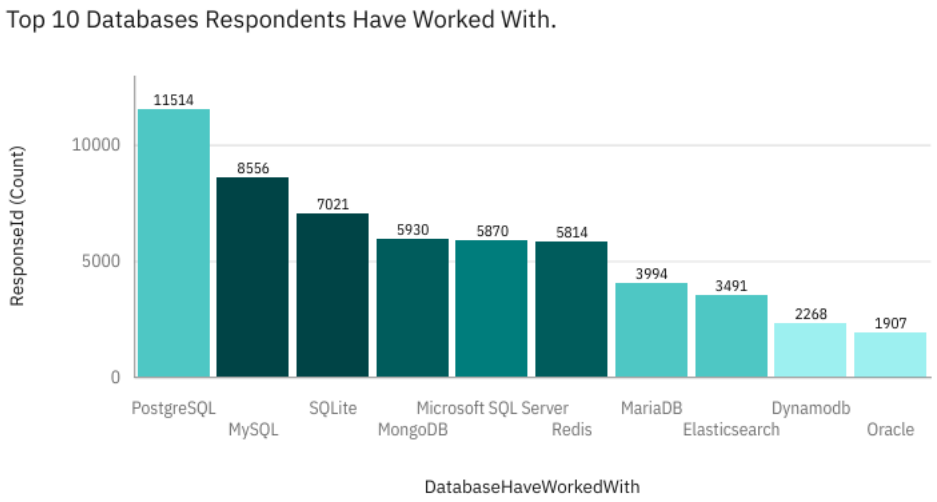
Implications

- Companies should prioritize JavaScript/TypeScript for frontend and Go/Rust for high-performance backend roles to attract talent.
- Recruitment for legacy languages like PHP may become increasingly difficult as developer interest wanes.

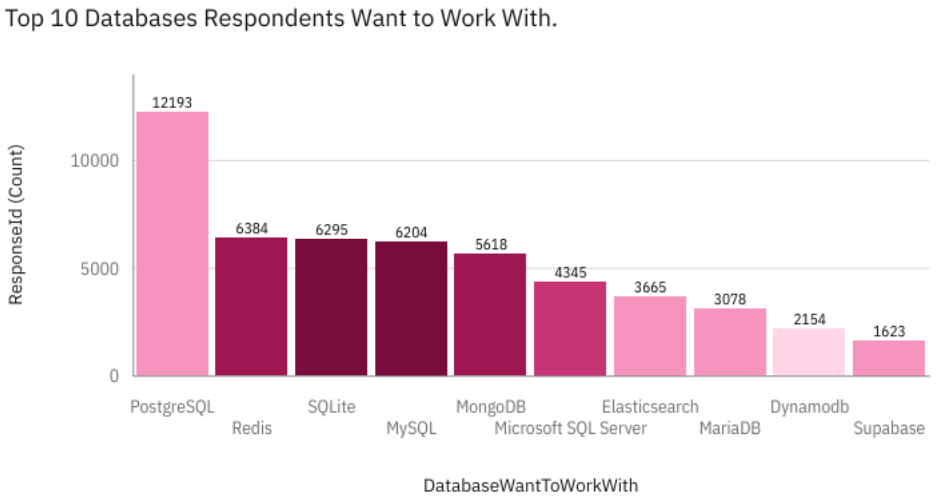
DATABASE TRENDS

Summarize key trends shown in the charts.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL leads both current use and future desire.
- Redis jumps from 6th in usage to 2nd in future interest.
- MySQL usage is higher than its future demand.

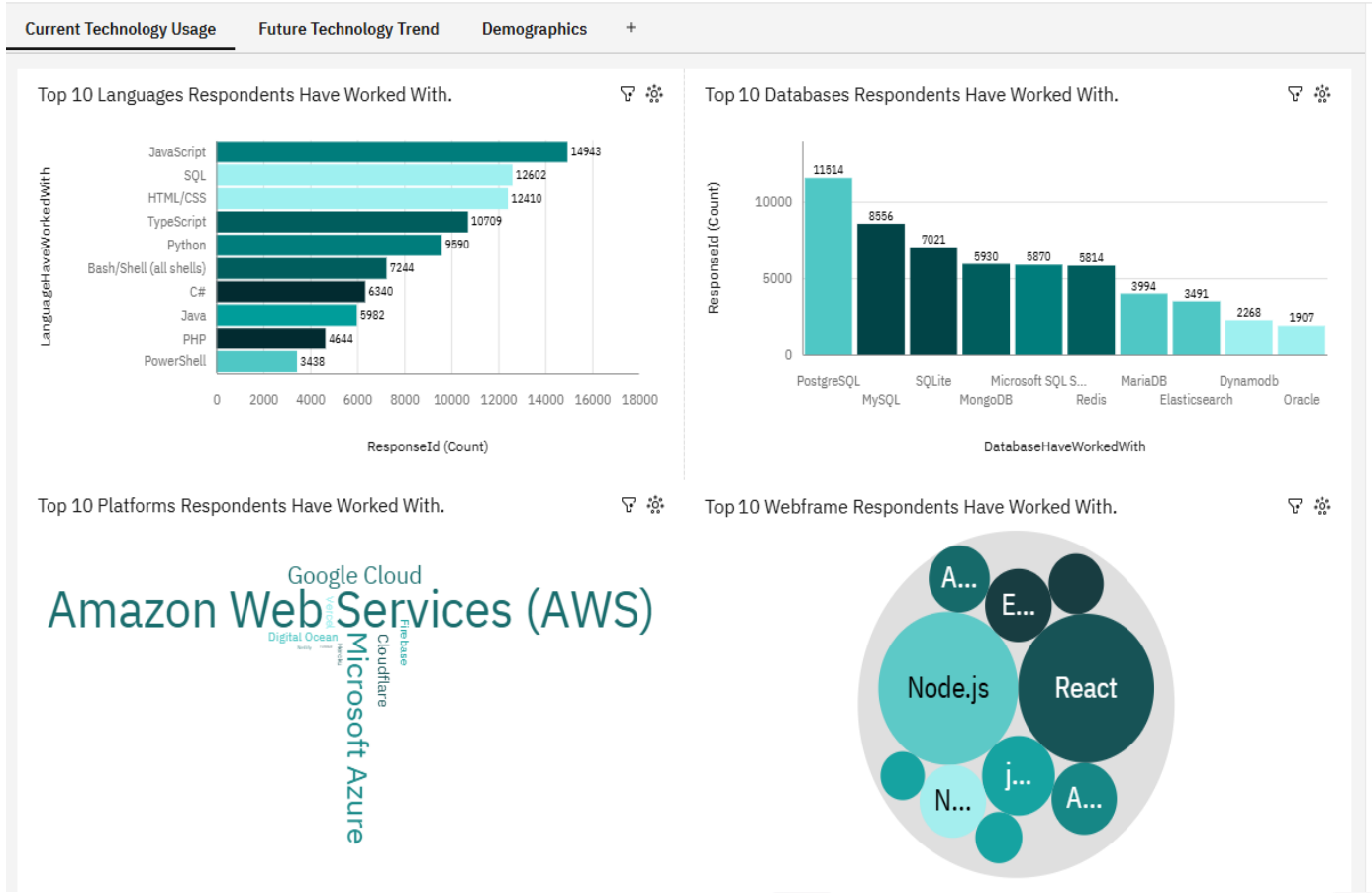
Implications

- PostgreSQL is the industry standard; infrastructure investments here are low-risk.
- The surge in Redis interest suggests a growing need for high-speed, in-memory data solutions in modern architectures.

DASHBOARD

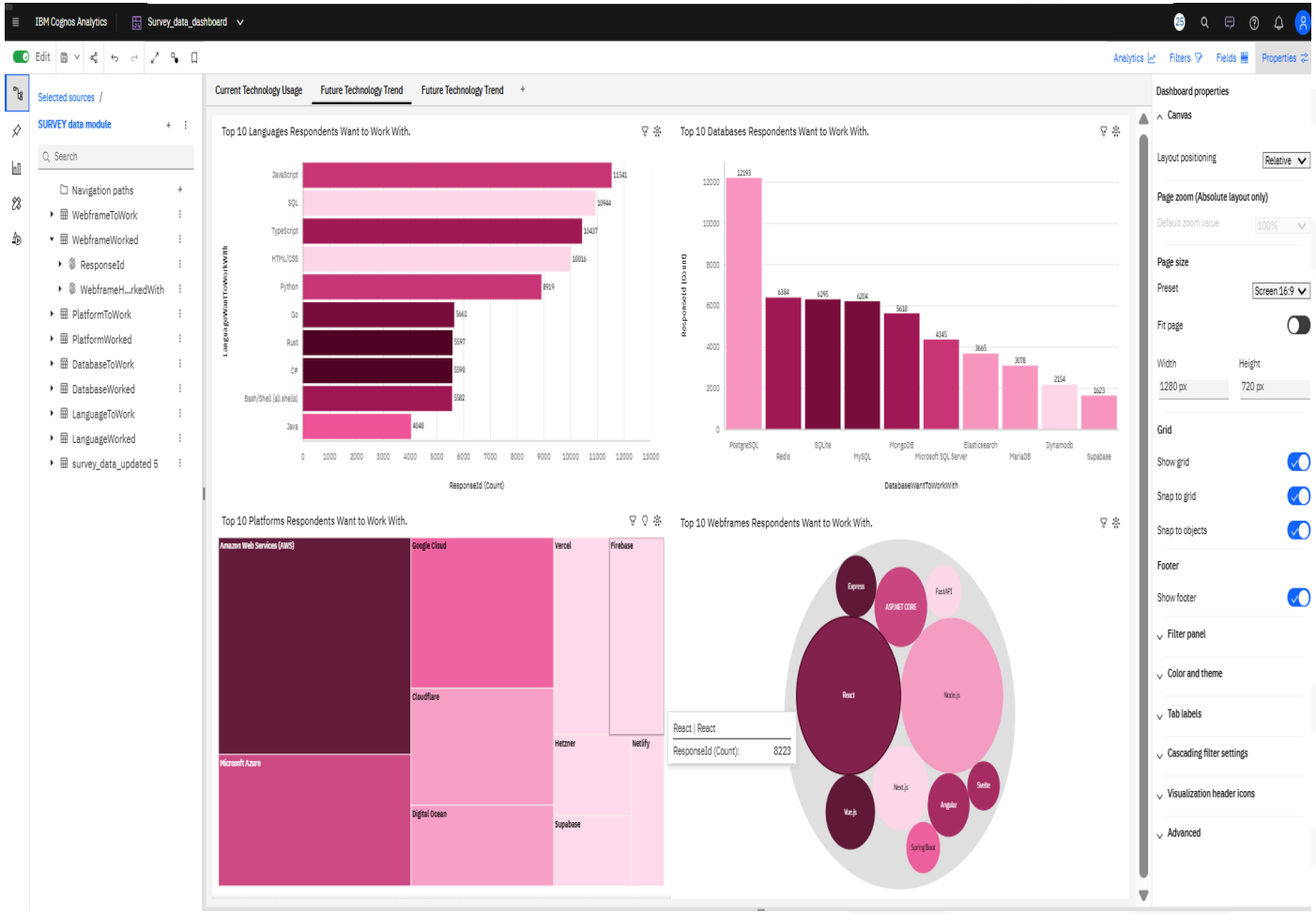
<Please present your dashboard in the following slides.>

DASHBOARD TAB 1: Current Technology Usage



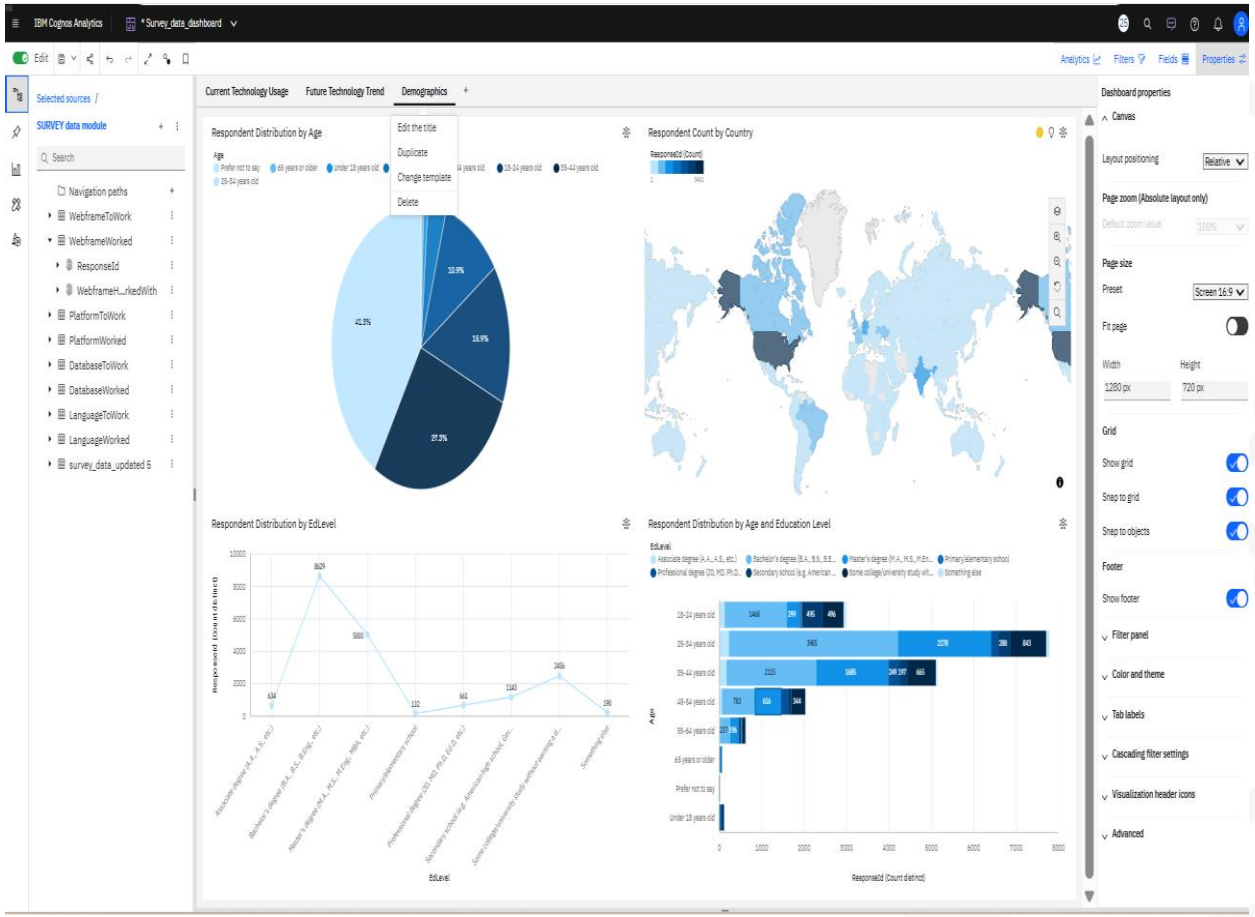
- The top Language currently used is Javascript with 14943 respondents.
- PostgreSQL, MySQL and SQLite are the top databases used.
- AWS is the most used platform followed by Ms. Azure.
- Node.js and React are the most used web frames respectively.

DASHBOARD TAB 2: Future Technology Trends



- Javascript still tops the top languages respondents want to use/learn.
- Postgre SQL is the database most respondents want to work with.
- AWS and MS. Azure are still the top platforms respondents want to work with.
- React and Node.js are the top web frames respondents want to work with respectively.

DASHBOARD TAB 3: Demographics



- Most respondents in this survey data are between 25-34 years.
- Majority of the respondents are from the USA(3441)
- A Bachelor’s Degree is the most common qualification, followed by Master’s Degrees among respondents.
- The 25–34 and 35–44 age groups show the highest density of both Bachelor’s and Master’s degrees.

DISCUSSION

- Interest is shifting towards Go and Rust for future projects languages.
- Redis shows significant upward momentum in rising database interest.
- The majority of the workforce is between 25-44 years.
- Respondents aged 18–24 make up 15.9% of the survey, indicating a steady influx of younger talent into the industry.
- PostgreSQL is the top database in both current usage and future interest.
- React and Node.js are the primary frameworks developers are currently working with and plan to use in the future.
- JavaScript is the most used and most admired programming language.

OVERALL FINDINGS & IMPLICATIONS

Findings

1. JavaScript is top used and desired language. There’s interest in Go and Rust for future projects.
2. PostgreSQL leads both current use and future desire.
3. React and Node.js dominate the trend bubble chart for future work .
4. AWS and Google Cloud are current leaders . Supabase and Vercel appear emerging interest for future work.
5. 41.3% of respondents are 25–34 years old .

Implications

1. The company should prioritize JavaScript, Go and Rust to attract talent.
2. PostgreSQL is the industry standard; infrastructure investments here are low-risk.
3. The web frameworks is consolidating around React-based stacks. Focus should be on these to ensure long-term project maintainability.
4. AWS remains the enterprise choice.
5. The workforce is academically credentialed in its career "prime" (25–44).

CONCLUSION

- The analysis confirms that JavaScript, SQL, and PostgreSQL remain the foundational pillars of the current developer landscape. There’s also a strategic shift toward Go and Rust.
- There’s a shift data management preferences. Future demand is pivoting toward PostgreSQL and Redis.
- The web development ecosystem is consolidating around React, Node.js, and Next.js.
- The primary technical workforce is highly educated and concentrated in the 25–34 age bracket.
- To remain competitive, the firm should prioritize recruitment and internal upskilling in Cloud-native technologies (AWS/Vercel) and Systems languages (Rust/Go). Aligning training portals with these high-growth "desired" skills will ensure the consultant pool stays ahead of market demand.

APPENDIX

- Primary dataset: Stack Overflow Developer Survey (2026)
- Tools and libraries: Python (Pandas, NumPy, Matplotlib, Seaborn), data collection(BeautifulSoup, Selenium, Requests API) and visualizations (IBM Cognos Analytics, Jupyter Notebooks).